

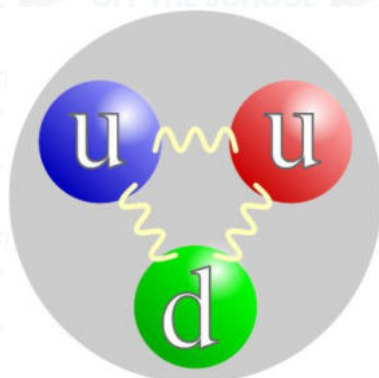


PARTICLE PHYSICS



Particle Physics is the branch of modern Physics that deals with the study and search of the ultimate constituent of matter and their interactions.

The standard model classified all known elementary particles into two main classes.



Fermions:

These are the matter particles, which make up everything in the universe.

Bosons:

These are the force-carrying particles, which mediate the interactions between fermions.

The Standard Model is a very successful theory. It has been able to predict the existence of many new particles, and it has been confirmed by a wide range of experiments.

However, the Standard Model is not perfect. It does not explain gravity, which is the fourth fundamental force.

It also does not explain dark matter and dark energy, which make up most of the universe.

Fundamental forces and their field particles:

The term "field particles" refers to particles associated with force fields.

According to the Standard Model, bosons are often considered field particles because they are linked to force fields.

For example, the photon is a field particle associated with the electromagnetic field.

Scientists have grouped all fundamental forces into four basic types. In order of increasing strength, these forces and their associated field particles are described as:

1. Gravitational Force:

It is the force that attracts two masses towards each other. It is the weakest of the four fundamental forces but acts over long distances.

Field Particle: The hypothetical particle associated with gravity is the graviton, although it has not been observed yet. The Standard Model doesn't explain gravity.

2. Weak Nuclear Force:

The weak force is responsible for radioactive decay, where unstable atomic nuclei break down into smaller, more stable nuclei. It's responsible for processes like beta decay and neutrino emission. This force is weaker than electromagnetic and strong nuclear forces but stronger than gravitational force.

Field Particle: There are three field particles associated with weak nuclear force. These are W^+ , W^- and Z bosons. These short-lived bosons carry the force over very small distances, explaining the limited range of the weak force.

3. Electromagnetic Force:

This force is responsible for the interactions between charged particles, such as electrons and protons. It includes both electric and magnetic forces. The electromagnetic force is stronger than both gravitational and weak nuclear forces. It is also a long-range force, similar to the gravitational force.

Field Particle: The field particle of electromagnetic force is photon. It is mass less and charge less particle. When ever charged particles interact, they exchange photons, causing the attractive or repulsive forces we observe.

4. Strong Nuclear Force:

The strong force binds quarks together to form protons and neutrons, and it holds atomic nuclei together. It is the strongest force among all forces and acts at subatomic levels.

Field Particle: Gluons are the field particles that mediate the strong force between quarks.

Particle Zoo:

One way of studying elementary particles is to classify them into different categories based on certain behaviours and then to look for similarities or common characteristics among the classifications.

We know that the Standard Model has classified all elementary particles and their interactions into two main groups according to their spins:

- Particles that carry force, called boson, having spin in integer values such as 0, 1, and 2.
- Particles that make up all matter, called fermions, having spin in odd half-integer values such as $1/2$, $3/2$, etc.

Bosons:

The arrangement of all elementary particles in the Standard Model constitutes a particle zoo.

All bosons can be classified into two types:

- Elementary Boson
- Composite Boson

Composite bosons consist of quark and anti-quark combinations, while elementary bosons are not made up of other elementary particles.

Elementary Bosons:

Elementary bosons are the carriers of the fundamental forces in nature. For examples:

- Gluon
- W^+ , W^- and Z^0 Bosons
- Photon
- Higgs Boson – gives mass to all particles.
- Graviton

The properties of elementary bosons are given in the table:

Force	Boson	Spin	Strength	Mass
Strong	Gluon	1	1	Massless
Electromagnetic	Photon	1	10^2	Massless

Weak	W, Z	1	10^7	80.91 GeV
Gravity	Graviton	2	10^{-39}	Massless

Composite Bosons:

Composite bosons are made up of an even number of fermions.

For example, mesons are composite bosons that consist of one quark and one antiquark.

Mesons are intermediate mass particles that mediate the strong force between nucleons such as protons and neutrons.

Fermions:

All material particles are made up of fermions. We can further divide fermions into two sub-classes:

- Elementary fermions
- Composite fermions

Elementary Fermions:

Elementary fermions are building blocks of all material particles. They are not made of any other particles. The fermions come in two types: leptons and quarks.

(a) Leptons:

Leptons are a group of elementary particles that do not experience the strong nuclear force.

Properties of leptons:

- **Electron:** Negatively charged; commonly found in atoms.
- **Muon and Tau:** Heavier counter parts of the electron.
- **Neutrinos:** Electrically neutral; they interact very weakly with matter.
- **Leptons** interact via weak and electromagnetic forces but not through the strong force.
- **Leptons** are stable particles and do not undergo decay under normal circumstances.
- **Leptons** exist alone and do not form groups.

Quarks:

Quarks are elementary particles that experience all three fundamental forces: strong nuclear force, weak nuclear force, and electromagnetic force.

Properties of Quarks:

- They carry fractional electric charges, either $+2/3$ or $-1/3$.
- Quarks interact strongly with the strong nuclear force, which is mediated by gluons. Quarks are never found as free particles in nature; they are always confined within larger particles called hadrons.
- Hadrons are particles made up of quarks held together by the strong nuclear force. Examples of hadrons include protons and neutrons.
- Quarks can undergo weak interactions, leading to processes such as beta decay. Weak interactions can change one type of quark into another. For example, a down quark can change into an up quark through weak decay processes.

Composite Fermions:

Composite subatomic particles such as protons, neutrons, alpha particles, etc., are composed of two or more elementary particles.

- All composite particles are massive.
- The composite particles that are made of quarks are called hadrons.

The two main categories of hadrons are baryons and mesons.

(a) Baryons:

Baryons are a class of hadrons that consist of three quarks. For example, protons and neutrons are the most well-known baryons, each composed of three quarks:

• Proton:

It consists of two up and one down quarks (uud). Up (u) quark has $+2/3$ charge and down quark has $-1/3$ charge.

Therefore, net charge on proton has $+1$ charge.

• Neutron:

It consists of three quarks: one up and two down quarks (udd). The net charge on neutron is zero.

(b) Mesons:

Mesons are another class of hadrons, but they consist of one quark and one anti-quark. For example, pions (π) are common mesons. They carry the strong nuclear force, binding protons and neutrons together.

• Quarks and Beta Decay:

- β^- -decay is a type of radioactive decay process in which a neutron in an atomic nucleus is transformed into a proton. When a β^- -decay occurs, a down quark emits a field particle called a W-boson (a weak force carrying particle) and transforms into an up quark.

- The W-boson quickly breaks up into an electron and an antineutrino.

The process is shown below:

- Electric charge is conserved during beta decay with the emission of an electron and an antineutrino.

- β^+ -decay is a type of radioactive decay process in which a proton is transformed into a neutron. In β^+ -decay, an up quark emits a field particle called a W^+ boson and transforms into a down quark. The W^+ boson quickly breaks up into an anti-electron (positron) and a neutrino. The process is shown below:

• Radiation Detectors:

- Radiation Detectors or Particle Detectors are devices that detect, track, and identify ionizing particles produced by nuclear decay, cosmic radiation, or particle accelerator reactions.

• Wilson Cloud Chambers:

- A Wilson cloud chamber is a type of tracking detector that works on the principle of ionization.



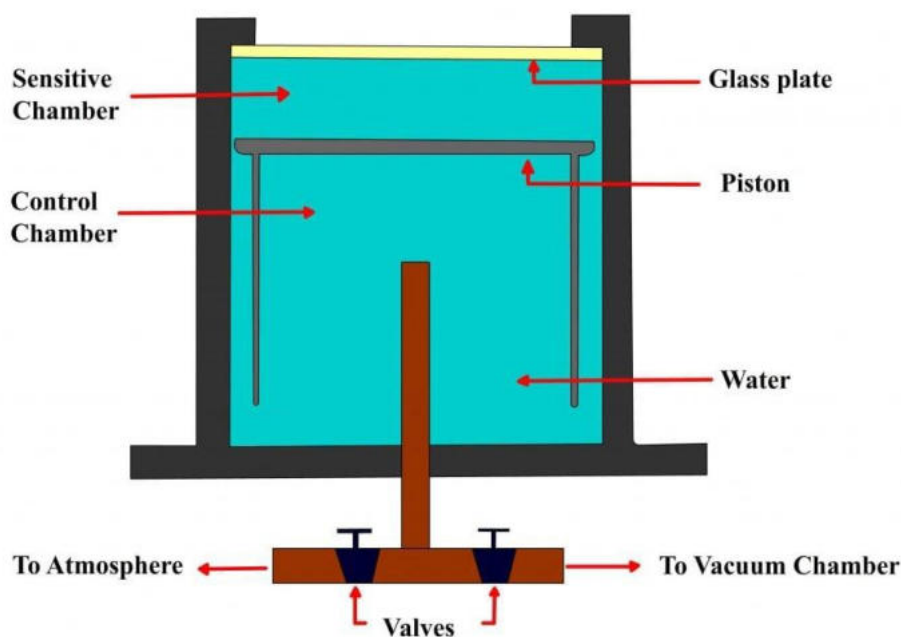
• Working Principle:

• The Wilson Cloud Chamber consists of a sealed container filled with a supersaturated vapor, typically water or ethanol. When a particle passes through the chamber, it ionizes the vapor, creating a trail of droplets that condense around the ionized path. This creates a visible cloud-like track that can be photographed and analyzed.

• Construction:

• The schematic diagram of Wilson cloud chamber is shown in the figure, which consists of a large cylindrical chamber A, with walls and a ceiling made of glass. It contains dust-free air saturated with water vapor. P is a piston working inside the chamber. When the piston moves down rapidly, adiabatic expansion of the air inside the chamber takes place.

The piston is connected to a large evacuated vessel F through a valve V. When the valve is opened, the air under the piston rushes into the evacuated vessel F, thereby causing the piston to drop suddenly. The wooden blocks WW reduce the air space inside the piston. Water at the bottom of the apparatus ensures saturation in the chamber. The expansion ratio can be adjusted by altering the height of the piston.



• As soon as the gas in the expansion chamber is subjected to sudden expansion, the ionizing particles are shot into the chamber through a slide window. A large number of extremely fine droplets are formed on all the ions produced by the ionizing particles. These droplets form a track of the moving ionizing particles. At this stage, the expansion chamber is profusely illuminated by a powerful beam of light L and two cameras CC are used to photograph the tracks as shown in figure 28.10.

The process of expansion, shooting of the ionizing particles into the expansion chamber, illuminating the chamber and clicking the camera must all be carried out in rapid succession in order to get satisfactory results.

• The type of ionizing particle can be identified by its track in the cloud chamber. Alpha particles, being relatively massive, travel in straight, thick, and clearly defined paths. Beta particles, however, produce less distinct tracks.

• There is a counter connected to this system to measure the radiation.

- **Working:**

- The chamber is filled with an inert gas (helium, neon, or argon) at low pressure. A high voltage is applied to this chamber. The metallic chamber will conduct electricity. When radiation enters the chamber through the window, the photons in the radiation will ionize the inert gas inside the chamber. This will make the gas conductive. The electrons produced due to ionization are accelerated due to the potential that we applied and these electrons cause even more ionization. The ionized electrons travel towards the anode. The anode is connected to a counter. The counter counts the electrons reaching the anode. This is how we measure radiation.

- **Use of GM Counter:**

- Geiger-Muller counters have a wide range of applications, including:

- **Radiation Monitoring:**

- They are commonly used for radiation monitoring in nuclear power plants, laboratories, and industrial settings to measure radiation levels and ensure the safety of workers and the environment.

- **Environmental Monitoring:**

- GM counters are employed for environmental radiation monitoring to assess background radiation levels and detect any abnormal increases in radiation.

- **Health Physics:**

- Health physicists use GM counters to monitor radiation exposure of individuals working with radioactive materials or in radiation-prone environments.

- **Education:**

- Health physicists use GM counters to monitor radiation exposure of individuals working with radioactive materials or in radiation-prone environments.

- **Radiological Emergencies:**

- In the event of radiological emergencies or accidents involving radioactive materials, GM counters can be used to assess radiation contamination levels.